

Technology Transfer Contracts in Vietnam: Mapping Regulatory Gaps through Topic Modelling and Doctrinal Analysis

Abstract

This article identifies regulatory gaps in Vietnam’s law governing technology transfer (TT) contracts by combining computational topic modelling of public discourse with doctrinal legal analysis. A corpus of thirty-four Vietnamese-language media and policy documents on science, technology and innovation was decomposed using Non-negative Matrix Factorisation ($K=7$), cross-validated against Latent Dirichlet Allocation and TF-IDF/LSA/BM25 clustering (Adjusted Rand Index = 0.448), and interpreted through Strauss-and-Corbin-style axial coding into four higher-order dimensions. These dimensions were then mapped onto the statutory architecture of TT contracts under Chapter III of the 2017 Law on Technology Transfer (Điều 21–34) and cross-checked against Vietnam’s existing statutory framework and the peer-reviewed legal literature. The central finding is methodological as much as substantive: the “essential facilities doctrine” theme, surfaced independently from the lexical structure of media discourse, coincides precisely with a gap identified in published Vietnamese scholarship on the interface between intellectual property and competition law — namely, the absence of a case-by-case balancing test capable of distinguishing a legitimate refusal to license from an abuse of market dominance. Building on this convergence, the article systematises seven categories of regulatory gaps in the current TT framework — spanning restrictive licensing terms, technology valuation, state-funded intellectual property ownership, damages for volatile technology assets, digital-technology security screening, and strategic-technology classification — and identifies, for each, the extent to which the Law on Technology Transfer must be read together with the Competition Law, the Intellectual Property Law, the Law on Prices, and public asset management legislation. The article closes with five sets of institutional recommendations for consideration as Vietnam implements the amended Law on Technology Transfer (Law No. 115/2025/QH15) and the new Law on Science, Technology and Innovation (Law No. 93/2025/QH15).

Keywords: technology transfer contract; topic modelling; axial coding; essential facilities doctrine; regulatory gap; Law on Technology Transfer; Vietnam.

1 Introduction

Technology transfer occupies a central place in Vietnam’s science-, technology- and innovation-led development strategy, as articulated in Resolution No. 57-NQ/TW of the Politburo (22 December 2024) on breakthroughs in science, technology, innovation and national digital transformation, and in the associated resolutions and strategies that have followed. The statutory framework governing this field is, at the same time, undergoing substantial reform: the 2017 Law on Technology Transfer (Law No. 07/2017/QH14) has been

amended by Law No. 115/2025/QH15 (expected to take effect on 1 April 2026), the Law on Science, Technology and Innovation (Law No. 93/2025/QH15) has replaced the earlier framework law on science and technology, and a series of implementing decrees and circulars has been issued through 2025–2026.

Vietnamese legal scholarship on TT contracts is well developed along several distinct lines: general assessments of the statutory framework (Hồ Minh Khánh, 2024); the interface between intellectual property and competition law (Bùi Thị Hằng Nga, 2019); patent commercialisation by public research institutes and universities (Phan Quoc Nguyen & To Nhat Mai, 2024; Phùng Minh Hải, 2026); and intellectual property valuation (Nguyễn Thị Minh Phương & Phí Công Thường, 2026). These contributions, however, typically proceed independently of what contemporaneous public discourse — news reporting, policy documents, expert commentary — reveals about the technology market at any given moment. Conversely, computational analyses of media and policy text tend to stop at describing which technologies are attracting attention, without engaging systematically with the statutory structure that governs the underlying contractual relationships. The two bodies of knowledge are therefore rarely triangulated against one another.

This article originates in an axial-coding study of a thirty-four-document media and policy corpus on science, technology and innovation in Vietnam, built on a topic model that has itself been cross-validated by several independent methods. Three research questions follow: first, where in the statutory structure of a TT contract do the themes that a computational model extracts from public discourse belong; second, at each such point, whether the current legal framework — the Law on Technology Transfer and the statutes with which it interacts — provides adequate regulation, or must instead be read together with other legislation; and third, what these gaps imply for the ongoing programme of legislative reform.

To answer these questions, the article draws on three sources: first, the topic-modelling and axial-coding results described above; second, a direct reading of the 2017 Law on Technology Transfer (Chapter III, “Technology Transfer Contracts,” Điều 21–34) together with an eighteen-article TT contract template in current use; and third, the published legal literature and the technology-transfer and patent disputes that have entered the public record through reported judgments or professional legal publications.

The article’s principal contribution is methodological: it demonstrates that an independent computational finding — a theme surfaced by topic modelling of media discourse — can converge with an independent finding of doctrinal legal scholarship, each corroborating the other, while offering a systematic map between the technology themes that dominate public discourse and the statutory architecture of the TT contract itself — a tool of potential use to both legislators and transactional counsel.

2 Literature Review And Methodology

2.1 Literature Review

The Vietnamese legal literature on TT contracts falls broadly into four strands. The first offers a general assessment of the statutory framework. Hồ Minh Khánh (2024), writing in the *Tạp chí Quản lý nhà nước* (State Management Review), observes that the 2017 Law on Technology Transfer contains no formal statutory definition of a “technology transfer contract,” proceeding instead by enumerating mandatory content requirements; the author further proposes importing a good-faith principle analogous to the Japanese doc-

trine of good faith and trust to strengthen contractual certainty, and notes that inter-agency coordination on technology assessment during investment licensing remains unclear, creating a risk that obsolete technology enters the country undetected.

The second strand examines the interface between intellectual property and competition law. Bùi Thị Hằng Nga (2019), in the National Assembly’s *Tạp chí Nghiên cứu Lập pháp* (Legislative Studies Review, No. 13(389)), analyses the essential facilities doctrine as a jurisprudential basis on which an intellectual property holder may be compelled to license where the protected asset constitutes an irreplaceable condition of market entry, surveying the relevant United States (*MCI v. AT&T*) and European Union (*Magill*; *Microsoft*) case law together with Vietnamese practice. Critically, the author cautions against treating a refusal to license as a *per se* infringement whenever the rights-holder occupies a dominant position, arguing that such rigidity fails to account for the value of patent exclusivity as an incentive to invest in innovation, and recommends instead a case-by-case balancing test that weighs competitive harm against the incentive effects of exclusivity.

The third strand concerns patent commercialisation from the public research sector. Phan Quoc Nguyen and To Nhat Mai (2024), writing in *Edelweiss Applied Science and Technology*, report that Vietnamese universities filed 192 patent applications in 2023 but obtained only fifty grants, and that technology-transfer revenue accounts for only two to five percent of total university income; drawing on the Bayh-Dole framework in the United States and comparable models in Japan and China, they argue for stronger university technology-transfer offices. In earlier work published in the *US-China Law Review*, Phan Quoc Nguyen (2020) records 10,250 technology transfer contracts concluded between 2001 and 2006, while Vietnamese universities accounted for only around four percent of domestically granted patents, with more than ninety percent held by foreign entities — evidence of a substantial gap between public-sector research capacity and commercialisation capacity. Phùng Minh Hải (2026), in the *Vietnam Journal of Science and Technology* (Series A), analyses the “bottlenecks” in commercialisation decisions at research institutes and universities, spanning accounting recognition, valuation and financing arrangements. From a comparative policy perspective, Lê Hồng Hiệp (2026), writing for *Nghiên cứu quốc tế* (International Studies), contrasts the United States’ Bayh-Dole model with the South Korean experience — where the commercialisation rate reached only around twenty-six percent, against roughly forty-one percent in the United States, despite South Korea’s having replicated the American statutory framework — cautioning that Vietnam risks a similar shortfall if patent ownership is vested in universities without a corresponding reform of research evaluation and reward structures toward commercialisation outcomes rather than publication and filing counts.

The fourth strand addresses intellectual property valuation. Nguyễn Thị Minh Phương and Phí Công Thường (2026), in the *Vietnam Journal of Science and Technology*, identify five specific gaps in the current legal framework for valuing and appraising intellectual property assets in Vietnam: the absence of detailed valuation principles and criteria; an underdeveloped market for the sale and assignment of intellectual property assets, which starves valuation of comparable transaction data; the absence of a national database recording licensing and capital-contribution transactions involving intellectual property; the absence of rules addressing valuation where no comparable asset exists; and the absence of training and professional-practice standards for intellectual property valuers.

This article positions itself as a complementary layer to these four strands. Rather than pursuing a single

doctrinal question in depth, it uses topic modelling to observe the TT contract’s entire statutory structure through the lens of public discourse, and then maps each finding onto the corresponding body of published scholarship. The approach follows the “computational grounded theory” tradition (Nelson, 2020), which uses computational tools to extend, rather than substitute for, the qualitative researcher’s theoretical judgment and the existing doctrinal literature.

2.2 Data and Methods

Corpus. The corpus comprises thirty-four Vietnamese-language media and policy documents on science, technology and innovation, drawn from public news reporting and policy sources and assigned identifying codes (e.g., HDCG_A01, Bài14) to preserve traceability throughout the analysis.

Topic modelling. Non-negative Matrix Factorisation (NMF) was applied with $K=7$ topics, a value selected on grounds of interpretability together with clustering stability. To test the robustness of the seven-topic structure, the analysis was cross-validated against three independent methods: Latent Dirichlet Allocation (LDA; Blei et al., 2003), KMeans clustering on a TF-IDF/LSA representation, and BM25-similarity-based clustering. Agreement between NMF and LDA was measured by the Adjusted Rand Index (ARI): $ARI(NMF, LDA \text{ at } K=7) = 0.448$, substantially higher than $ARI(NMF, LDA \text{ at an automatically selected } K=3) = 0.060$ — indicating that the seven-topic structure is considerably more stable than a coarser structure, and is not an artefact of any single algorithm.

Axial coding. The seven second-order topics (NMF-1 through NMF-7) were interpreted qualitatively using the Strauss and Corbin (1998) paradigm model — causal conditions, the core phenomenon, context, intervening conditions, action/interactional strategies, and consequences — and consolidated into four higher-order (third-order) dimensions reflecting the broader axes of concern within public discourse on science, technology and innovation.

Doctrinal cross-checking. For each higher-order dimension, the corresponding location within the statutory structure of a TT contract was identified by direct reference to the 2017 Law on Technology Transfer (Chapter III, Điều 21–34, read in full) and to an eighteen-article TT contract template currently in use, and the extent of any regulatory gap at that point was assessed against the published legal scholarship identified in Section 2.1 and against publicly reported TT and patent disputes.

Methodological limitations. First, a thirty-four-document corpus is modest by the standards of large-scale topic-modelling research and is exploratory rather than statistically generalisable in character; the percentages reported in Table 1 illustrate the relative structure of this particular corpus rather than the weight of these issues in Vietnamese technology policy discourse as a whole. Second, the statutory material analysed directly — the 2017 Law on Technology Transfer and the contract template — constitutes primary source material read in full; the provisions of Law No. 115/2025/QH15 and Law No. 93/2025/QH15, not yet in force as of the time of writing (entry into force expected 1 April 2026), are discussed at the level of publicly announced policy direction rather than as settled statutory text. Third, the disputes discussed in Section 4 were accessed through reliable secondary sources — published judgments and the professional publications of practising intellectual property counsel — rather than through direct access to the underlying case files, and are cited accordingly.

3 Findings

3.1 The Statutory Framework for TT Contracts as a Reference Structure

Chapter III (Điều 21–34) of the 2017 Law on Technology Transfer is devoted to TT contracts. Điều 22 requires that a TT contract be concluded in writing or an equivalent form, in a language agreed by the parties, and provides that the Civil Code, the Commercial Law, the Intellectual Property Law and the Competition Law apply concurrently where relevant. Điều 23 sets out fourteen mandatory content requirements: the name of the technology; the subject matter of the technology and the products it generates, together with applicable standards and quality specifications; the assignment of ownership or the licensing of use rights; the mode of transfer; the rights and obligations of the parties; the price and payment method; the term and effective date of the contract; definitions of terms used; the implementation schedule, timeline and place of performance; warranty obligations; liquidated damages for breach; liability for breach; the forum for dispute resolution; and any other matter agreed by the parties. Điều 24 governs the term and effectiveness of the contract; Điều 25 and Điều 26 set out the respective rights and obligations of the transferor and the transferee; Điều 27 governs price and payment method (five permissible payment methods, together with a mandatory audit requirement where state capital is involved); Điều 28 governs technologies subject to restricted transfer, which require a Technology Transfer Licence; and Điều 29–32 govern the procedure for contract registration and licensing.

This statutory structure is reflected almost exactly in contract-drafting practice: the eighteen-article TT contract template examined in this study (Article 1, “Subject Matter of the Transfer,” through Article 18, “Implementation Provisions”) tracks the sequence of mandatory content set out in Điều 23 closely. The practical significance of complete compliance with these formal and substantive requirements is more than theoretical. In a case decided on appeal by the High People’s Court in Da Nang (Bản án số 04/2019/KDTM-PT, 22 January 2019), the consequences of an incomplete patent transfer agreement were laid bare: the parties had produced only a two-page document that omitted any provision on price or payment method, leaving the court without an adequate basis to determine the agreement’s validity or the parties’ specific rights and obligations, and the first-instance judgment was set aside for retrial. The case illustrates that the mandatory content items listed in Điều 23 — often treated in practice as a matter of drafting formality — in fact play a decisive role in protecting the parties’ interests once a dispute arises.

3.2 Mapping the Four Higher-Order Dimensions onto the TT Contract Structure

Table 1 sets out the mapping between the four higher-order dimensions derived from topic modelling and the statutory structure of a TT contract. The “Adequacy of Current Law” column reflects a qualitative assessment based on the scholarship reviewed in Section 2.1 and on direct reading of the statutory text.

Table 1: Mapping the four higher-order dimensions (from topic modelling) onto the TT contract structure under the 2017 Law on Technology Transfer

Higher-order dimension (weight)	Constituent second-order topics	Location in the TT contract	Adequacy of current law
1. Foundational TT institutions and practice (58.8%; n=20/34)	NMF-1, general TT activity (n=13); NMF-2, innovation/education (n=4); NMF-6, essential facilities doctrine (n=3)	Điều 22 (formation, form), Điều 23 (mandatory content), Điều 25 (transferor's rights and obligations, including restrictive terms binding the transferee)	Uneven: the formation/content framework (Điều 22–23) is comparatively developed; restrictive licensing terms (linked to NMF-6, see 3.3) lack a case-by-case balancing mechanism, as warned by Bùi Thị Hằng Nga (2019)
2. Digital technology applications (14.7%; n=5/34)	NMF-4: industrial AI, energy, data	Điều 23(2) (subject matter/products), Điều 28 (restricted/prohibited technology list)	Incomplete coverage: Điều 28 and its accompanying list predate the wave of generative AI and the data economy; screening criteria for digital technology remain unspecified
3. Strategic national technology (14.7%; n=5/34, proposed merger)	NMF-5: semiconductors (n=3); NMF-7: quantum computing, cybersecurity (n=2)	Điều 28 (restricted-transfer list), policy direction under Resolution No. 57-NQ/TW	Under development but not yet codified: “strategic technology” (semiconductors, quantum) currently exists chiefly at the level of policy instruments, not yet translated into classification criteria under Điều 28
4. Agricultural technology (11.8%; n=4/34)	NMF-3: agriculture, AI	Điều 23(2), Điều 27 (price — particularly the valuation of agricultural technology involving biological or seed-related elements)	No dedicated valuation regime: no valuation criteria exist for biotechnology/agricultural technology; parties must rely on a general valuation framework that itself remains underdeveloped (Nguyễn Thị Minh Phương & Phí Công Thường, 2026)

Note: Dimension 3 reflects a proposed restructuring — merging NMF-5 (semiconductors) and NMF-7 (strategic technology: quantum, cybersecurity) into a single “strategic national technology” dimension, rather than grouping NMF-7 with NMF-4 as in the initial clustering, because the two topics share a common policy logic (export control and core-technology security) distinct from the “industrial application” logic of NMF-4.

3.3 The Central Finding: Convergence between Topic Modelling and Published Legal Scholarship on the Essential Facilities Doctrine

The study’s most methodologically significant finding is the convergence of two independent sources of knowledge at topic NMF-6. This topic was extracted by the NMF model purely from the lexical features of three corpus documents (HDCG_A01, HDCG_A09, HDCG_A16) — no legal knowledge was supplied to the model a priori — and was interpreted, during axial coding, as reflecting the essential facilities doctrine: a situation in which an undertaking holding an irreplaceable, essential technology refuses to transfer or license it to a competitor, thereby entrenching a dominant market position.

Independent cross-checking against Điều 22 of the 2017 Law on Technology Transfer — which applies both the Intellectual Property Law and the Competition Law to TT contracts — and against Bùi Thị Hằng Nga’s (2019) study in the *Tạp chí Nghiên cứu Lập pháp* reveals that this is precisely the point of intersection that Vietnamese legal scholarship has already flagged as lacking an adequate evaluative framework. Specifically, that study finds that the prevailing approach in practice — treating a refusal to license as a per se infringement whenever the rights-holder occupies a dominant position — is unduly rigid, failing to reflect the value of patent exclusivity as an incentive for investment in innovation, and recommends a case-by-case evaluative mechanism that weighs competitive impact against innovation incentives simultaneously. In other words, a mathematical model operating purely on lexical features “rediscovered” — from an entirely different vantage point, grounded in media discourse rather than statutory text — precisely the complex point of legal intersection that dedicated scholarship, using comparative case-law analysis, had already identified. This two-directional convergence — one data-driven, the other theory-driven — is the basis on which this study identifies the finding as its central contribution, examined further in Section 4.1.

4 Regulatory Gaps And Reliance On Other Statutes

Building on the mapping in Section 3 and on the published scholarship reviewed in Section 2.1, this section systematises seven specific gaps in the current TT legal framework, identifying at each point which other statute effectively bears the regulatory burden that the 2017 Law on Technology Transfer does not itself discharge.

4.1 The Absence of a Balancing Mechanism for Restrictive Licensing Terms: Reliance on the Competition Law and Compulsory Licensing under the Intellectual Property Law

This is the article’s central finding (Section 3.3). Điều 25 of the 2017 Law on Technology Transfer sets out the transferor’s rights and obligations, while TT contracts in practice frequently include terms restricting the transferee — prohibitions on accessing competing technology, limitations on the scope of use, or tying obligations requiring the purchase of inputs or components from the transferor. The 2017 Law on Technology

Transfer establishes no mechanism for distinguishing a restrictive term that is “reasonable and necessary to protect the technology” from one that constitutes an abuse of a dominant market position. Bùi Thị Hằng Nga’s (2019) research shows that this gap has two faces. On one hand, the 2018 Competition Law and the compulsory-licensing mechanism under the Intellectual Property Law — which permits the State to authorise a third party to use a patented invention without the owner’s consent where doing so is “necessary to meet an urgent need of society” — are the principal tools available to address a refusal to transfer essential technology. On the other hand, applying these tools on the assumption that every refusal to license amounts to a per se infringement risks undermining the very investment incentives that intellectual property law is designed to protect. The 2017 Law on Technology Transfer itself offers no guidance for reconciling these two values within the contract, nor does it establish any explicit coordination mechanism between the competition authority and the science-and-technology regulator when such disputes arise — an institutional gap that leaves a party refused access to an essential technology to choose, unassisted, between a competition-law claim and an application for compulsory licensing.

4.2 Absent Valuation Criteria and Data: Reliance on the 2023 Law on Prices

Điều 23(6) and Điều 27 of the 2017 Law on Technology Transfer treat price and payment method as mandatory contract content, listing five permissible payment methods and requiring a mandatory audit where state capital is involved, but the law prescribes no specific principle or criterion for valuing technology. Nguyễn Thị Minh Phương and Phí Công Thường (2026), writing in the *Vietnam Journal of Science and Technology*, systematise five specific gaps in the current legal framework for valuing and appraising intellectual property: the absence of detailed valuation principles and criteria, producing inconsistency across cases; an underdeveloped market for the sale and assignment of intellectual property assets, which leaves valuation without comparable transaction data; the absence of a national database recording licensing transactions and capital contributions in kind involving intellectual property; the absence of rules for valuation where no comparable asset exists; and the absence of training and professional-practice standards for intellectual property valuers. These gaps in the 2023 Law on Prices are reflected directly in Điều 27 of the Law on Technology Transfer, leaving the price term in TT contracts — particularly where state capital is involved — frequently without a robust quantitative basis, and creating a natural risk-aversion on the part of state-capital representatives who must approve a transfer price.

4.3 The Boundaries of Ownership of State-Funded Intellectual Property: Public Asset Management Law

For technology and inventions generated through state-funded science-and-technology tasks — common among public universities and research institutes, a sector that, according to Phan Quoc Nguyen (2020), accounted for only around four percent of domestically granted patents despite thousands of TT contracts concluded between 2001 and 2006 — the determination of ownership and of the authority to license or commercialise is governed concurrently by public asset management legislation, which was designed primarily for conventional public assets (land, premises, vehicles) rather than for intellectual property. Recent amendments to the Intellectual Property Law have begun to address part of this problem: according to specialist financial press reporting, the amendments establish that patents generated with state funding may be assigned only to domestic organisations or individuals, with the amendments to the Law on Science and Technology

and to public asset management legislation revised in parallel to preserve consistency. A more concrete step has been piloted under Decree No. 271/2025/NĐ-CP on the commercialisation of scientific research results within Hanoi City, which permits public science-and-technology institutions in the city to establish spin-off enterprises to commercialise research outputs — a notable mechanism, though currently a local pilot rather than a nationally applicable rule.

4.4 A Damages Regime Poorly Suited to Highly Volatile Technology Assets

Paragraphs 11–12 of Điều 23 of the 2017 Law on Technology Transfer treat liquidated damages and liability for breach as mandatory contract content, but as a matter of principle these provisions apply the Civil Code 2015’s general damages regime — actual, provable loss — which sits uneasily with the character of technology assets, whose loss from a given breach (for example, disclosure of confidential technology or unauthorised onward transfer to a third party) is often impossible to quantify at the moment of breach and can fluctuate substantially with subsequent market developments. The Intellectual Property Law offers a more flexible mechanism for damages arising from infringement of intellectual property rights (permitting a statutory damages award within prescribed limits where actual loss cannot be proven), but this mechanism applies only where the technology is protected as an established intellectual property object (a patent, a trade secret, and so forth); where transferred technology is not tied to a protection title, a TT contract must fall back on the Civil Code’s actual-damages regime — a gap for which the 2017 Law on Technology Transfer currently offers no dedicated solution.

4.5 Incomplete Security Screening for Digital Technology and AI under the Restricted-Transfer List

Điều 28 of the 2017 Law on Technology Transfer and its accompanying restricted-transfer list were built on a conventional technology paradigm (chemical, biological, and civil-military dual-use technology) and do not yet cover the non-traditional security risks posed by emerging digital technologies: dual-use artificial-intelligence models, large-scale training data, and high-performance computing infrastructure. This is the gap corresponding to higher-order Dimension 2 in the topic-modelling mapping (Section 3.2): topic NMF-4 (industrial AI, energy, data) has no clear statutory anchor in the current control framework beyond the general, unelaborated language of Điều 28. Vietnamese TT law currently addresses such cases chiefly through the general dual-use export-control framework — not specific to digital technology — and, in part, through cybersecurity legislation, another instance in which a gap must be filled by other law, while dedicated frameworks for this field remain under development in many jurisdictions and no settled model yet exists for reference.

4.6 “Strategic National Technology” (Semiconductors, Quantum Computing) Not Yet Codified as a Classification Criterion

Corresponding to higher-order Dimension 3 in the topic-modelling mapping (Section 3.2) — the proposed merger of NMF-5 (semiconductors) and NMF-7 (quantum computing, cybersecurity) — the concept of “strategic technology” currently exists chiefly at the level of policy instruments. Resolution No. 57-NQ/TW identifies quantum technology as one of Vietnam’s strategic priorities, and the relevant authorities are developing a dedicated strategy for the field, but the concept of “strategic technology” has not yet been codified

as a classification criterion under Điều 28 of the Law on Technology Transfer or the 2014 Law on High Technology. As a result, the specific incentives and restrictions that policy envisages for semiconductor and quantum technology — which require a distinct legal regime combining investment incentives with control over outbound transfer — currently lack a statutory basis for coherent implementation, pending the implementing decrees and circulars expected under the new Law on Science, Technology and Innovation (Law No. 93/2025/QH15) and the amended Law on Technology Transfer (Law No. 115/2025/QH15).

4.7 Registration and Licensing Procedures: Administratively Burdensome, though Reform Is Under Way

Điều 29–32 of the 2017 Law on Technology Transfer govern the registration of TT contracts and the licensing of restricted-transfer technology. The regulator itself has acknowledged the underlying problem: according to a decision on administrative procedures issued by the Ministry of Science and Technology, reported by the Government’s online newspaper on 9 January 2026, a series of processing timeframes has been shortened — the review period for a TT contract registration certificate reduced from three to two working days, the period for constituting an advisory council reduced from five to three working days, and the period for issuing a decision following the council’s report reduced from five to two working days, alongside the introduction of an online filing channel through the National Public Service Portal. This reform indicates that the regulator itself regarded the prior process as unduly protracted; at the same time, the persistent gap between the number of patents granted and the number of TT contracts formally registered — reflected in the modest contribution of the public research sector to both figures (Phan Quoc Nguyen, 2020) — suggests that the practical effectiveness of the registration mechanism remains a matter to be monitored following the 2026 procedural reform.

Table 2: Summary of seven regulatory gaps and the statutes/institutions relied upon to fill them

#	Gap	Related provisions of the 2017 Law on Technology Transfer	Statute/institution relied upon
1	No balancing mechanism for restrictive licensing terms	Điều 22, Điều 25	2018 Competition Law; compulsory licensing under the Intellectual Property Law
2	No valuation criteria or data for technology pricing	Điều 23(6), Điều 27	2023 Law on Prices (itself incomplete as to IP valuation)
3	Unclear ownership boundaries for state-funded intellectual property	(indirect, via the transferor's identity)	Public asset management legislation; Law on Science and Technology
4	Damages regime unsuited to highly volatile technology assets	Điều 23(12)	Civil Code 2015; Intellectual Property Law (only where a protection title exists)
5	Incomplete security screening for digital technology/AI	Điều 28 and accompanying list	General dual-use export-control framework; cybersecurity legislation
6	“Strategic national technology” not yet a classification criterion	Điều 28	2014 Law on High Technology; Resolution No. 57-NQ/TW; Law No. 93/2025/QH15
7	Registration/licensing procedures administratively burdensome (reform under way)	Điều 29–32	(an implementation gap, not a normative one)

5 Discussion

5.1 Comparison with Published Doctrinal Literature

Each of the four strands of scholarship surveyed in Section 2.1 — the general assessment of the statutory framework (Hồ Minh Khánh, 2024), the intellectual property/competition-law interface (Bùi Thị Hằng Nga, 2019), public-sector patent commercialisation (Phan Quoc Nguyen & To Nhat Mai, 2024; Phan Quoc Nguyen, 2020; Phùng Minh Hải, 2026; Lê Hồng Hiệp, 2026), and intellectual property valuation (Nguyễn Thị Minh Phương & Phí Công Thường, 2026) — pursues a discrete question in depth, using doctrinal analysis, comparative law, or policy analysis. This article’s contribution is broader but shallower than any single one of these studies: it surveys the entire statutory structure of the TT contract rather than one dimension of it, and cross-references that structure against public policy discourse, but does not itself undertake case-law research or independent field interviews. Its contribution is accordingly one of synthesis and cross-validation rather than of novel doctrinal discovery: it shows that the gaps identified independently by discrete strands of specialist research — restrictive licensing terms, valuation, public asset ownership, commercialisation — can be located consistently within a single statutory map of the TT contract, and that they also surface independently in public discourse.

5.2 Methodological Contribution

The article’s principal methodological contribution is to illustrate a three-layer triangulation process: first, within the topic-modelling exercise itself (NMF cross-validated against LDA and alternative clustering methods, measured by ARI); second, between the computational finding and doctrinal analysis of the primary statutory text (Section 3.3); and third, between the article’s findings and the independent, previously published scholarship reviewed in Sections 2.1 and 4. This approach follows the “computational grounded theory” tradition (Nelson, 2020), which uses computational tools to extend the qualitative researcher’s field of observation rather than to substitute for theoretical judgment. In legal-policy research on a relatively young field such as technology transfer, this approach may assist in identifying emerging themes — such as Dimension 3, strategic national technology — before they are fully reflected in statutory text, thereby supporting proactive rather than purely reactive lawmaking.

5.3 Limitations

Three limitations bear on the interpretation of these findings. First, a thirty-four-document corpus is small; the percentages reported in Table 1 illustrate the relative structure of themes within this particular corpus, not an estimate of the weight of these issues within Vietnamese technology-policy discourse as a whole. Second, the disputes discussed in Sections 3.1 and 4 were accessed through reliable secondary sources — published judgments and the professional publications of practising counsel — rather than through the underlying case files, and the number of such disputes cited remains limited, insufficient to generalise about adjudicative trends. Third, the discussion in Section 4.6 of “strategic national technology” is necessarily more predictive of policy direction than analytical of settled statutory text, since Law No. 115/2025/QH15 and Law No. 93/2025/QH15 had not yet entered into force at the time of writing; the relevant assessments should be revisited once the implementing instruments are issued in full.

6 Conclusion And Recommendations

This article has presented an analysis combining topic modelling of media and policy discourse with an examination of Vietnam’s current legal framework for TT contracts, cross-checked against the published legal literature. The findings show that the four higher-order dimensions derived from the data map reasonably clearly onto the Điều 21–34 structure of the 2017 Law on Technology Transfer, and that at each point of correspondence there exists some degree of regulatory gap that the current TT framework must address by reference to other statutes — the Competition Law, the Intellectual Property Law, the Law on Prices, public asset management legislation, and the Law on High Technology. The most salient finding — the convergence between the “essential facilities doctrine” theme, surfaced independently by a computational model, and the gap in balancing mechanisms for restrictive licensing terms identified by published legal scholarship — demonstrates the value of this combined approach.

On the basis of the seven regulatory gaps identified in Section 4, the article proposes five sets of institutional recommendations for consideration as Vietnam develops the implementing instruments for the amended Law on Technology Transfer (Law No. 115/2025/QH15) and the new Law on Science, Technology and Innovation (Law No. 93/2025/QH15).

First, Điều 25 of the Law on Technology Transfer, or its implementing regulations, should be supplemented with a case-by-case evaluative framework for distinguishing “reasonable” restrictive terms from those amounting to an abuse of dominance — along the lines recommended by Bùi Thị Hằng Nga (2019) — together with an explicit coordination mechanism between the competition authority and the science-and-technology regulator for disputes involving the refusal to transfer essential technology.

Second, Vietnam should accelerate the development of dedicated principles and criteria for intellectual property valuation under the 2023 Law on Prices and establish a national database of licensing transactions and in-kind capital contributions involving intellectual property, as recommended by Nguyễn Thị Minh Phương and Phí Công Thường (2026), while amending Điều 27 of the Law on Technology Transfer to encourage market-indexed pricing mechanisms — revenue-based royalties, milestone payments — rather than confining parties to a single lump-sum payment structure.

Third, the rules governing ownership and exploitation of state-funded intellectual property should continue to be harmonised across the Intellectual Property Law, the Law on Science and Technology, and public asset management legislation, and the spin-off enterprise mechanism piloted under Decree No. 271/2025/NĐ-CP — currently confined to Hanoi City — should be evaluated for selective nationwide extension once the pilot has been assessed.

Fourth, the restricted- and prohibited-transfer technology list under Điều 28 should be reviewed and updated to capture emerging digital technologies (dual-use AI models, large-scale data infrastructure), and the concept of “strategic national technology” (semiconductors, quantum computing) should be codified as a concrete classification criterion, rather than remaining at the level of policy instruments alone.

Fifth, the effectiveness of the TT contract registration and licensing reforms implemented from early 2026 should continue to be monitored against concrete indicators — the number of contracts registered, and that

number as a proportion of total patents granted — to ensure that faster processing times translate into a genuine increase in formal TT contract registration.

Future research should extend the media and policy corpus to test the robustness of the four higher-order dimensions on a larger sample, and should undertake independent field research — surveys and interviews — to verify directly the prevalence of the gaps identified here in the drafting and performance of TT contracts, particularly once the amended Law on Technology Transfer (Law No. 115/2025/QH15) formally enters into force on 1 April 2026.

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